Bhalaji Yadav Kantepalle

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PROFESSIONAL SUMMARY

Chemical Engineering Professional (M.S. Candidate) specializing in **Quantitative Metrology**, **Soft Matter Mechanics**, and **Process Automation**. Expertise lies in bridging the gap between R&D and Manufacturing—using tools like **Python (OpenCV)** and **Advanced Rheology** to solve complex material problems, while leveraging **SAP ERP** and **FDA Compliance** frameworks to ensure scalability. Proven track record of securing research funding (**$30k**), compressing operational timelines by **>90%**, and driving digital transformation in regulated environments.

EDUCATION

Virginia Commonwealth University (VCU) — Richmond, VA
*M.S., Chemical & Life Science Engineering | **GPA: 3.62*** — *Exp. Dec 2026*

Thesis: “Beyond Peel Strength: Stability-Driven Adhesion Metrics for Soft Wearable Interfaces”

Focus: Fracture Mechanics, Mechanochromic Sensors, Process Optimization, Experimental Design (DOE).

Achievement: Graduate Assistantship Award (Fall 2025) – Full Tuition Waiver & Stipend.

Amrita University — Coimbatore, India
B.Tech., Chemical Engineering — *Jun 2022*

TECHNICAL SKILLS

Materials Characterization: **Fracture Mechanics**, Adhesion Testing (ASTM D2724), Rheology (Cross/Power-law), Optical Microscopy
Computational R&D: **Python** (OpenCV, Pandas, SciPy, NumPy), **Docker**, Image Processing, **CIE L*a*b* Colorimetry**
Operational Excellence: **SAP ERP Validation** (OQ/PQ), **V-Model Framework**, Lean Manufacturing, RCA, CAPA
Regulatory Compliance: **FDA 21 CFR Part 11**, **cGMP** (ICH Q7 / US FDA), FDA Class I Mapping, GDP

RESEARCH EXPERIENCE

VCU Soft Functional Materials Lab — Richmond, VA
Graduate Researcher & Metrology Lead — *Sep 2024 – Present*

- **Novel Optical Metrology:** Engineered a high-throughput **Computer Vision pipeline** (Python/OpenCV) to quantify mechanochromic response in smart textiles. Synchronized sub-pixel texture tracking with **CIE L*a*b* colorimetry**, increasing metrology throughput by **300%**.
- **Fracture Mechanics Automation:** Developed a first-principles analysis protocol to identify "false positive" adhesion failures. Automated the extraction of **Peel Stability Index (PSI)** metrics, reducing data processing time by **>80%**.
- **Material Characterization:** Characterized viscoelastic properties of cholesteric LCs using **Cross and Power-law rheology models**. Correlated shear history to helical pitch using **Abbe Refractometry** to optimize optical sensitivity.
- **Grant Success ($30,000): Drafted** a successful research proposal for the Commonwealth Cyber Initiative (CCI CVN). **Substantiated** the clinical viability of thermochromic textiles through **market gap analysis** and **FDA Class I regulatory mapping**.

INDUSTRY EXPERIENCE

Kreative Organics Pvt. Ltd. — Hyderabad, India
Technical Project Manager — *May 2023 – May 2024*

- **SAP System Requalification:** Re-engineered the deployment strategy to meet a strict 24-hour executive deadline. Compressed the standard 5-day timeline into a **5-hour overnight operation** (**>90% reduction**), achieving zero production downtime.
- **FDA Compliance & Quality Systems:** Led the **Quality Assurance interface** for SAP ERP implementation. Translated manufacturing SOPs into digital validation protocols (**OQ**/**PQ**), ensuring system logic enforced **cGMP** safety controls.

Technical Project Intern — *Oct 2022 – Apr 2023*

- **Chemical Data Strategy: Spearheaded** the initiative to standardize CAS/IUPAC nomenclature across global trade databases. **Directed** the "Chemical Logic" algorithm and **engineered** Docker/Tableau workflows, reducing manual research time by **70%**.

PUBLICATIONS & ACHIEVEMENTS

Journal Article: I. Caloian, J. Trapp, **B. Y. Kantepalle**, et al. “Mechanical Properties of Dual-Layer Electrospun Fiber Mats.” *Polymers* 17(13), 1777 (2025). doi:10.3390/polym17131777

Selected Participant: 1st National Neutron Scattering School, Oak Ridge National Laboratory (Sep 2025).

Poster Presentation: “Adhesives for Personalized Wearable Devices” (ACS Fall 2025, Washington D.C.).